Multi-Domain and Reasoning-Effort Reinforcement Learning

Deep Dive into Kimi K3 — Chapter 21

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Foundation Books Project

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Goal: define nine domain–effort specialists, derive the strict-overrun override, compute real updates of a four-route policy, follow a max-to-low curriculum, and keep success, cost, and compliance apart before consolidation.

Roadmap

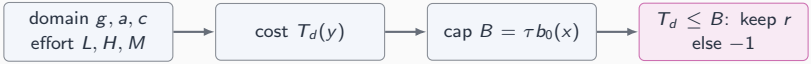
The budgeted reward

Learning under a cap

Curricula, evidence, and handoff

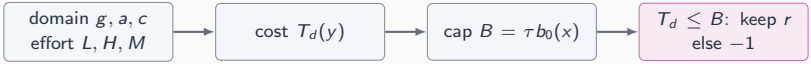
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Nine specialists and one override



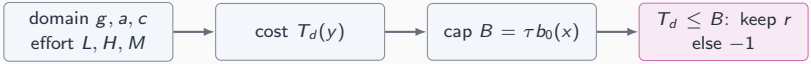
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- Calibration matters: if $\tau b_0(x) < c_\star(x)$, every success scores below a within-budget failure.

Learning under a cap

A four-route policy learns

- Routes A, B, C, D cost $40, 400, 120, 70$ and succeed $0.35, 0.95, 0.85, 0.75$; caps $500, 160, 80$. Gradient $\partial J_B/\partial z_i = p_i(q_i - J_B)$; first high-cap step from $J_0 = -0.095$.

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Effective reward can rise while accuracy falls; the objective rewards success inside the cap, not every token saved.

Curricula, evidence, and handoff

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- Max then high then low: at cap 80 the rare route D is still minor after 200 exact updates and dominant by 500; curriculum seeds cross at a median of 140 versus 287.5, at higher cumulative token cost. [OBSERVED: 30-seed finite study]

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Effort control is only as good as the task reward. Next: Chapter 22's Rewards, Verifiers, and Agentic Generative Reward Models.